Complex segments reduce computational complexity

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Slides available at
https://aphonologist.github.io/presentations

Complex segments in non-local harmony

- may not pattern like individual components
- $\widehat{\alpha\beta} \neq \alpha\ \beta$

Analyses with complex segments are less powerful

- TSL $\subsetneq$ I-TSL
- novel methodology to argue for complex segments

Phonotactics

English (eng, Indo-European)

- ✔ {…, ˈkʰæt, ləˈbubu, ˈpʰlo͡ɪk, …}*
- ✖ {…, ˈΘæt, ləbubu, ˈlpʰo͡ɪk, …}

Which set does a given string belong to?

How much computation does answering require?

ployk: A club or bludgeon (obsolete); https://doi.org/10.1093/OED/3430233413

Strings and tiers

Overwhelming majority of generalizations[†]

- finite set of banned / penalized strings

Some English generalizations

- no clicks: *[ˈʘæt]
- no initial son-obs clusters *[⋊ˈlpʰo͡ɪk⋉][‡]

Longest banned string has length *k*

- check all windows of length *k* for bad strings
- *strictly local*[§]

[†]Heinz (2018)
[‡]Rogers and Pullum (2011)
[§]McNaughton and Papert (1971)

Strings and tiers

Overwhelming majority of generalizations[†]

- finite set of banned / penalized strings

Another English generalization

- no stressless lexical words: *[ləbubu]
- cannot be modeled by banning xyz^k

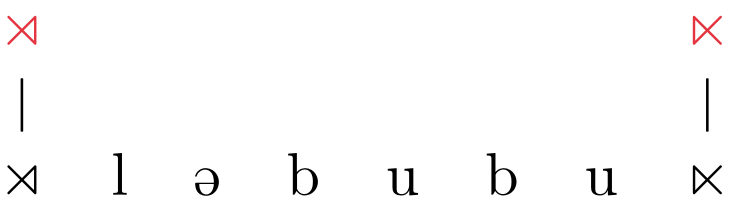
*[ləbububu. . . bububu]

[†]Heinz (2018)

Strings and tiers

Long-distance generalizations as strings on tiers

- tier for stress = {ˈ, ˌ}
- no empty tiers *⋊⋉

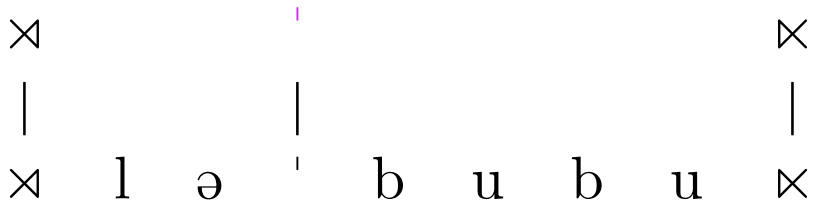


- *tier-based strictly local*†

†Heinz et al. (2011)

Strings and tiers

Long-distance generalizations as strings on tiers



- phonologically awkward
  $\sim$ every word must contain ŋ
  $\Rightarrow$ a theory of phonological syntax, not substance
- can also be stated over syllables†

†Heinz (2014); Baek (2017)

Strings and tiers

Long-distance generalizations as strings on tiers

- tier for stress = {ˈ, ˌ}
- cannot ban words with only secondary stresses

*[ˌ ˌ ˌ ˌ ˌ ˌ ˌ ˌ ˌ ˌ]

- cannot ban initial son-obs clusters

Different generalizations rely on different tiers

- *multi-tier-based strictly local*†
- even greater empirical adequacy

†McMullin (2016); Aksënova and Deshmukh (2018)

Main claim

Behavior of complex segments in harmony

- unusually complex if clusters
- typically simple if complex segments

Anterior harmony

Slovenian (slv, Indo-European; Jurgec, 2011:330-331)

- variable and dialect-specific

sl-ux	ʃl-iʃ-i	‘hearing/hears’
zapɔr	ʒapoɾ-niʃki	‘prison/-ADJ’
t͡sepət͡s	t͡ʃept͡ʃ-ək	‘fool/-DIM’
ʒival-i	ʒival-ski	‘animal-GEN/-ADJ’
t͡ʃel-o	t͡ʃel-ist	‘cello/cellist’

*[+anterior, +strident]…[−anterior, +strident]

Anterior harmony

Slovenian (slv, Indo-European; Jurgec, 2011:330-331)

- variable and dialect-specific

sit	nasit-iʃ	‘full / (you) feed’
zida	zida-ʃ	‘(s/he) builds / (you) build’
stoji	stoji-ʃ	‘stands / (you) stand’
t͡sitɾ-e	t͡sitɾ-aʃ	‘zither / zither player’

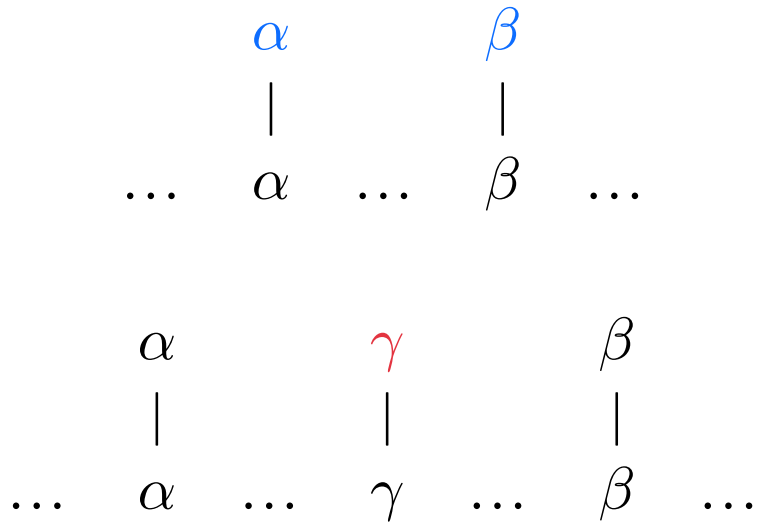
*[+anterior, +strident]…[−anterior, +strident]

- unless t, d intervene

Blocking

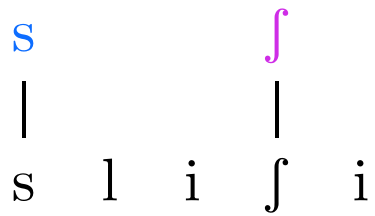
Blockers are benign projectees

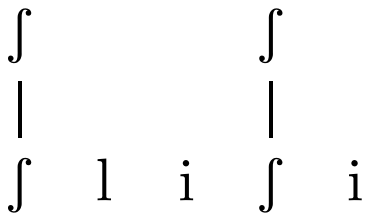
tier = {α, β, γ}; *$\alpha\beta$



An analysis

$$\text{tier} = \begin{bmatrix} -\text{sonorant} \\ +\text{coronal} \end{bmatrix}; \quad *\begin{bmatrix} +\text{anterior} \\ +\text{strident} \end{bmatrix} \begin{bmatrix} -\text{anterior} \\ +\text{strident} \end{bmatrix}$$





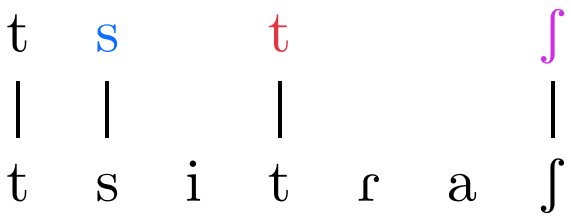
An analysis

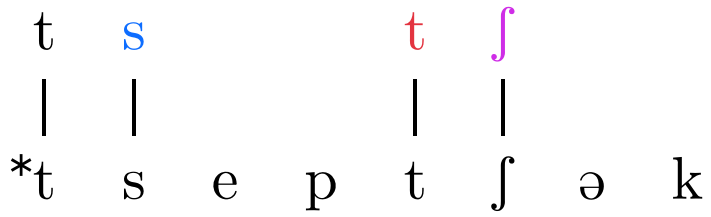
$$\text{tier} = \begin{bmatrix} -\text{sonorant} \\ +\text{coronal} \end{bmatrix}; \quad * \begin{bmatrix} +\text{anterior} \\ +\text{strident} \end{bmatrix} \begin{bmatrix} -\text{anterior} \\ +\text{strident} \end{bmatrix}$$

$$\begin{array}{ccccccc} \widehat{\text{ts}} & & \text{t} & & & \int & \\ | & & | & & & | & \\ \widehat{\text{ts}} & \text{i} & \text{t} & \text{ɾ} & \text{a} & \int & \end{array}$$

Affricates are not clusters

Adjacency lost on tier



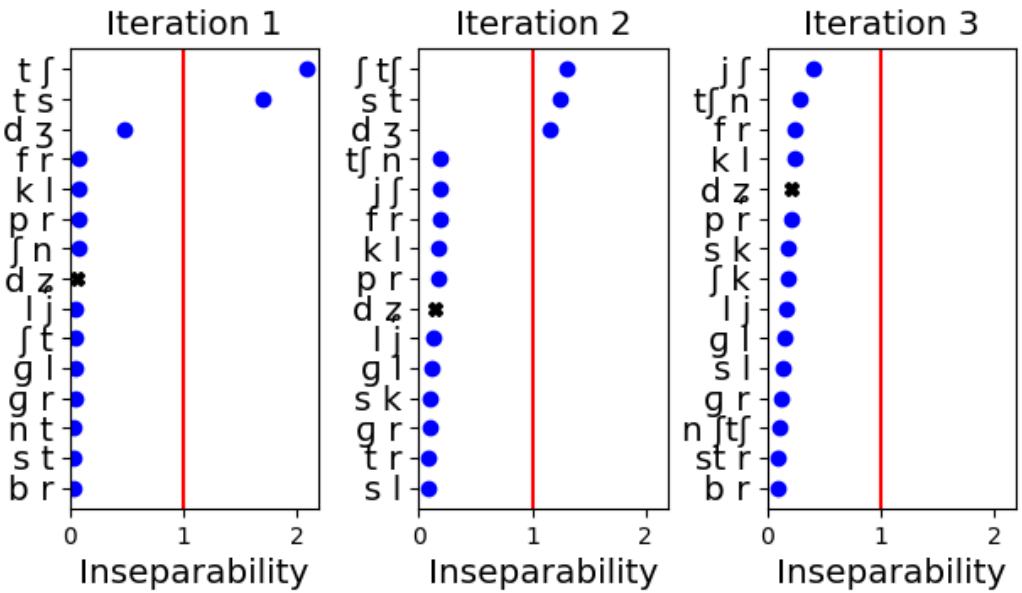


☠ t, d block except when followed by strident

Evidence for affricates

Jurgec (2011:330) explicitly includes them

Complex segment learner[‡] identifies them in Wiktionary data (>5,000 words)



[‡]Gouskova and Stanton (2021)

Nasal harmony

Yaka (yaf, Niger-Congo; Hyman, 1995:6, 9-10; Ruttenberg, 2000:67, 70)[§]

/kúd-idi/	[kúd-idi]	‘chasser qqn’ [chase sb]
/kún-idi/	[kún-ini]	‘planter’ [grow]
/mák-idi/	[mák-ini]	‘grimper’ [climb]
/nútúk-idi/	[nútúk-ini]	‘s’incliner’ [bow down]
/mí:tuk-idi/	[mí:tuk-ini]	‘bouder’ [to pout]

*[+nasal]…{d, l}

[§]Simplifying somewhat; see Lamont (2023)

Nasal harmony

Yaka (yaf, Niger-Congo; Hyman, 1995:6, 9-10; Ruttenberg, 2000:67, 70)[§]

/bíːm͡b-idi/	[bíːm͡b-idi]	‘embrasser’ [embrace]
/kúːn͡d-idi/	[kúːn͡d-idi]	‘enterrer’ [bury]
/táːŋ͡g-idi/	[táːŋ͡g-idi]	‘lire, compter’ [read, count]
/níːm͡b-idi/	[níːm͡b-ini]	‘dormir’ [sleep]
/njóːn͡dz-idi/	[njóːn͡dz-ene]	‘maigrir,tarir’ [lose weight,dry up]
/méːŋ͡g-idi/	[méːŋ͡g-ene]	‘haïr’ [hate]

*[+nasal]…{d, l}

- prenasalized stops/affricates are transparent

[§]Simplifying somewhat; see Lamont (2023)

Transparency

Transparent segments are not projected

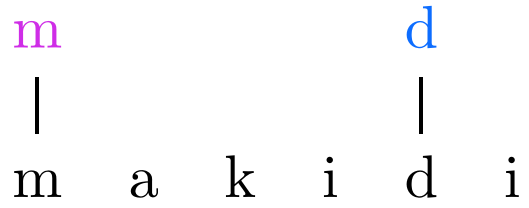
tier = {α, β}; *$\alpha\beta$

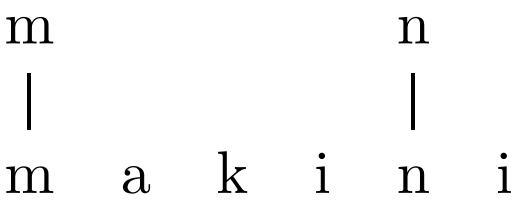
$$\begin{array}{ccccccc} & & \alpha & & \beta & & \\ & & | & & | & & \\ \ldots & & \alpha & \ldots & \beta & \ldots & \end{array}$$

$$\begin{array}{ccccccccc} & \alpha & & & & & \beta & & \\ & | & & & & & | & & \\ \ldots & \alpha & \ldots & \gamma & \ldots & & \beta & \ldots & \end{array}$$

An analysis

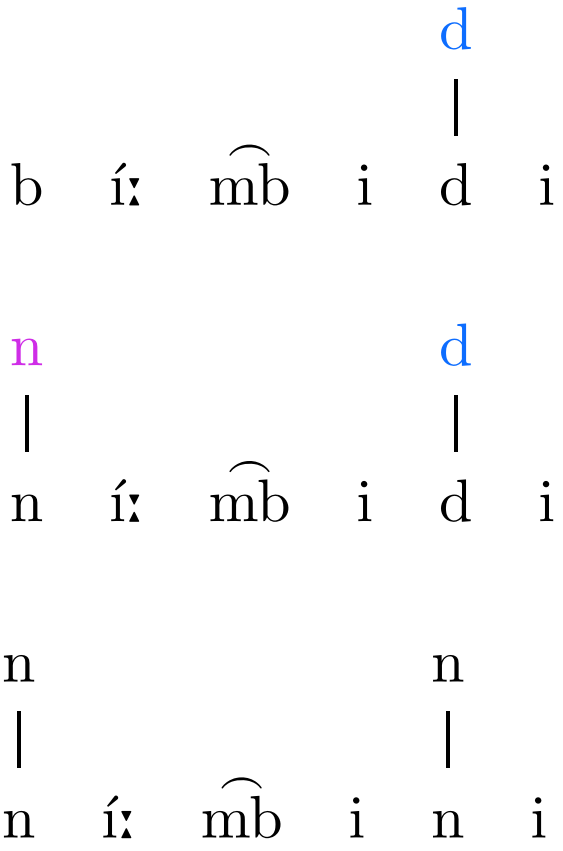
tier = [+nasal] ∪ {d, l}; *[+nasal][−nasal]





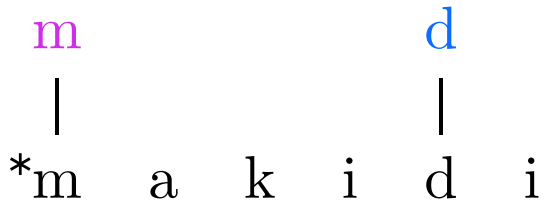
An analysis

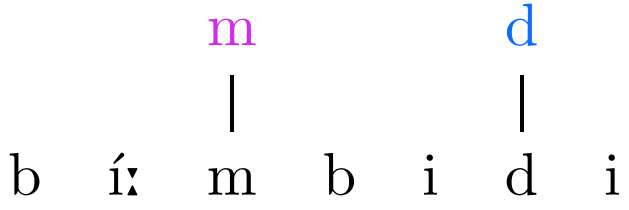
tier = [+nasal] ∪ {d, l}; *[+nasal][−nasal]



Prenasalized stops/affricates are not clusters

Adjacency lost on tier





☠ nasals trigger except when followed by stop

Evidence for prenasalized stops/affricates

Kidima (1991:3) and Hyman (1995) explicitly include them

Complex segment learner[¶] identifies them in Ruttenberg (2000)[‖]

[¶] Gouskova and Stanton (2021)

[‖] Thanks Larry Hyman for a digital copy!

Diphthongs in vowel harmony

Shigatse Tibetan (bod, Sino-Tibetan; Haller, 2012)

- height harmony
  *{i, y, ʊ, ə, i͡e}…{e, œ, ɔ, a}
  *{e, œ, ɔ, a}…{i, y, ʊ, ə, i͡e}
- i͡e patterns as a high vowel

Sakha (sah, Turkic; Sezer and Wetzels, 1986)

- high-mid diphthongs pattern as high vowels in round harmony

Maasai (mas, Nilo-Saharan; Cole and Trigo, 1988)

- glides [j, w] only participate in ATR harmony in diphthongs

Computational complexity

More power required without complex segments

- tier projection contingent on local context[**]

Local context can be defined phonologically → MTSL[††]

[**] De Santo and Graf (2019); Hao and Andersson (2019); Hao (2020); Johnson and De Santo (2024)

[††] Lamont (2023)

Interactions derive complexity

[anterior] harmony in Ineseño Chumash (inz, Chumashan; Applegate, 1972:10, 17, 119, 303)

[s̪am^{ʔ}ipus̪ at̪ɔk]	‘they call it tok (red milkweed); it is called tok’
[ʃχal̪amʃ]	‘it is wrapped’
[kʃɨpɨt̪waʃ]	‘I made acorn mush’
*[kʃɨpɨt̪was̪]	

Interactions derive complexity

[anterior] dissimilation in Ineseño Chumash (inz, Chumashan; Applegate, 1972:17, 117, 312, 337)

[ʃt̻ɛpuʔ]	‘he gambles’
[ʃn̻iʔ]	‘his neck’
[ʃl̻ɔk’in̻̥]	‘he cuts it’
[s̻ɛs̻qɛn̻it̻ t̻ik’ip]	‘he asks me what I said’
[s̻kut̻i̥]	‘he sees’
[ʔin̻is̻qil̻uχn̻ɨwɨn̻̥]	‘he never hurries’

Interactions derive complexity

Sibilant harmony overrides coronal dissimilation (Applegate, 1972:71, 120)

a.	/s̪-n̪ɛt̪-us̪/	[s̪-n̪ɛt̪-us̪]	‘he does it to him’
		*[ʃ-n̪ɛt̪-uʃ]	
		*[ʃ-n̪ɛt̪-us̪]	
b.	/s̪-iʃ-t̪iʃi-jɛp-us̪/	[s̪-is̪-t̪is̪i-jɛp-us̪]	‘they (dual) show him’
		*[ʃ-iʃ-t̪is̪i-jɛp-us̪]	
		*[ʃ-iʃ-t̪iʃi-jɛp-uʃ]	

+ lexical/variable exceptions (Heinz and Idsardi, 2010)

Interactions derive complexity

Sibilant harmony vs. coronal dissimilation is not TSL

- clearly derives from independently attested, conflicting generalizations[‡‡]
- conflict is powerful: SL vs. SL $\geq$ REG[§§]
- finite-state OT $\leq$ REG[¶¶]
- baseline / atomic generalizations still TSL

[‡‡] Heinz and Idsardi (2010) catalog a handful of these; see also Mayer and Major (2018) and Hampe (2026)

[§§] Koser and Jardine (2020)

[¶¶] Karttunen (1998)

Interactions derive complexity

I-TSL derives from phonological interaction

- not a privileged class like TSL
- are there any languages in I-TSL − TSL that are not decomposable? (and are not autosegmental)
  $/\mathrm{pi}\ldots\mathrm{du}/ \rightarrow [\mathrm{pi}\ldots\mathrm{du}]$
  $/\mathrm{\int a}\ldots\mathrm{du}/ \rightarrow [\mathrm{\int a}\ldots\mathrm{du}]$
  $/\mathrm{\int i}\ldots\mathrm{du}/ \rightarrow [\mathrm{\int i}\ldots\mathrm{di}]$
- maybe Lionnet (2016) has cases?

Easy as [abc]

Complex segments produce more phonologically natural analyses

- complexity classes relative to alphabet
- what is the alphabet? / what's a possible complex segment?
- not a question of formal complexity; a question of phonological substance

Easy as $[\mathrm{abc}]$

Complexity is relative to alphabet

- all regular languages reduce to strictly local[‡‡] encode transition function of FSA: $q_i = [\mathrm{O}^i]$
- natural language phonology is not a programming language
- "One could, for example, propose that phonological rules be finite transducers in the literal sense. No one would take such a suggestion seriously because of the linguistic inappropriateness of the formulations it would require." (Johnson, 1972:58)

[‡‡] Medvedev (1964)

Sums of their parts

Is it appropriate to equate $\widehat{\alpha\beta}$ and $\alpha\ \beta$?

A'ingae (con, isolate; Dąbkowski, 2025)

- optional labialization only targets diphthongs
  [fãʔfã] 'diarrhea'
  [fa͡e] ~ [fo͡e] 'one'
- representational analysis: [a] > [a͡]

In harmony, blocking/triggering effects dampened

- Unexpected (=find an attested case):
  [saka∫]
  [∫aka$\widehat{\text{t}}$∫]

Thank you!

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